

Abstract

Dairy Product Tracker

The Dairy Product Tracker is a web application built using the MERN stack (MongoDB, Express.js, React, Node.js) and Ethereum blockchain to ensure transparency and traceability in the dairy supply chain. It involves three main roles:

- Manufacturer – Adds product details
- Retailer – Marks the product as received and available
- Customer – Enters the product ID to verify product details

Why Ethereum?

Using Ethereum ensures openness and verifiability by anyone (like customers) without needing permission, making product information fully trustworthy.

Use of Blockchain:

It allows customers to easily check the authenticity and journey of dairy products from the manufacturer to the retailer. Ensures no one can fake expiry dates or product origin.

Decentralised Course Platform

This decentralized course platform enables Educators to register and upload courses with details like title, description, price, and content. Using Ethereum smart contracts, educators' ownership is securely recorded, and they automatically receive crypto payments when students enroll.

Students can browse courses, pay with ETH or test ETH, and gain permanent access, with proof of enrollment stored immutably on the blockchain.

Why Ethereum?

Ethereum's smart contracts automate payments and ownership verification without intermediaries, ensuring trust and transparency.

Use of Blockchain:

Blockchain provides a secure, tamper-proof ledger that records course ownership and student enrollments, guaranteeing permanent and verifiable access.

Photo Marketplace

It is a decentralized platform where Photographers can upload and list photos for sale, and Buyers can purchase photos by paying in ETH. Using Ethereum smart contracts, ownership transfers are securely recorded, and payments are automated.

Features:

- Photographers upload and list photos for sale
- Buyers browse and buy photos with ETH
- Automated, transparent payment and ownership transfer via smart contracts

Why Ethereum?

Ethereum allows photographers and buyers to securely buy and sell photos directly, with automatic payment and ownership transfer—no middleman needed—building trust between users.

Use of Blockchain:

Blockchain keeps a permanent, unchangeable record of each photo's ownership and sale, ensuring buyers receive authentic photos and preventing any fake or duplicated sales.